MATERIALS 2

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ECAM
SCHOOL OF ENGINEERING
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Course Description

- 3 Lectures, 3 Tutorials and 3 LABs

- Prerequisite knowledge: Chemistry, material 1

- Grading procedure:

10% Class assessment (quizzes, attendance, homework,…)
15% LAB work
30% Midterm exam
45% Final exam

MATERIALS 2

Lecture 1: Diffusion

Part 1: https://www.youtube.com/watch?v=btwrYz_XzIs&t=2s
Part 2: https://www.youtube.com/watch?v=gNhNb9kvDY8&t=4s

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Contents

Introduction

Reactions and processes that are important in the **treatment** of **material surfaces**

↓

depend on the **transfer of mass** either within a specific solid or from a liquid, a gas, or another solid phase

↓

This can be accomplished by diffusion

↓

phenomenon of **material transport** by **atomic motion**

Definitions

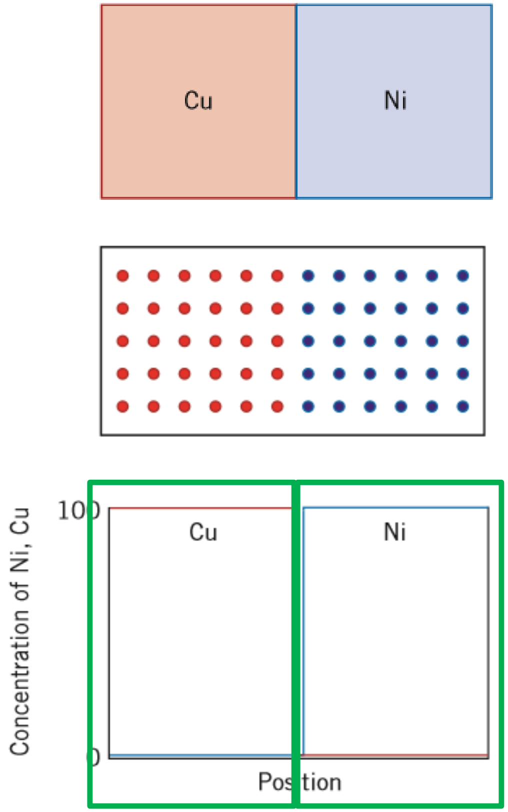
- **Diffusion**: a phenomenon that is demonstrated using a diffusion couple, which is formed by joining bars of two different metals together so that there is intimate contact between the two faces.

- **Self-diffusion**: is atomic migration in pure metals--i.e., when all atoms exchanging positions are of the same type.

- **Interdiffusion**: is the diffusion of atoms of one metal into another metal.

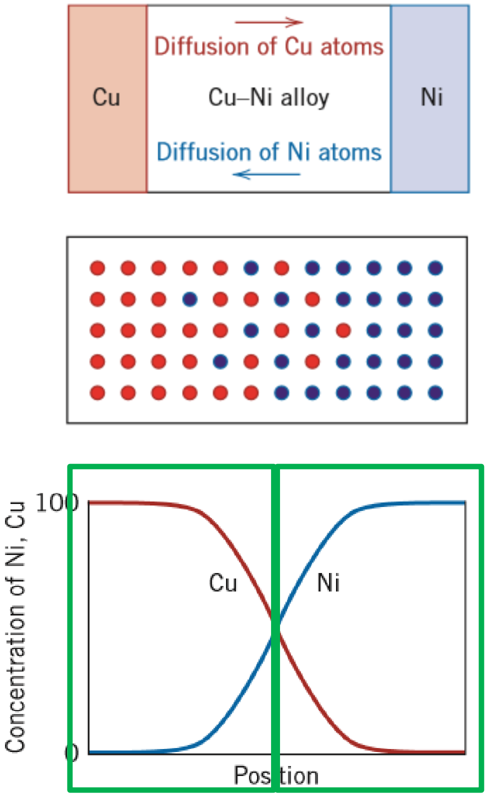
Example of Interdiffusion

- **Copper-nickel diffusion couple**



Heat

Treatment



Diffusion Mechanisms

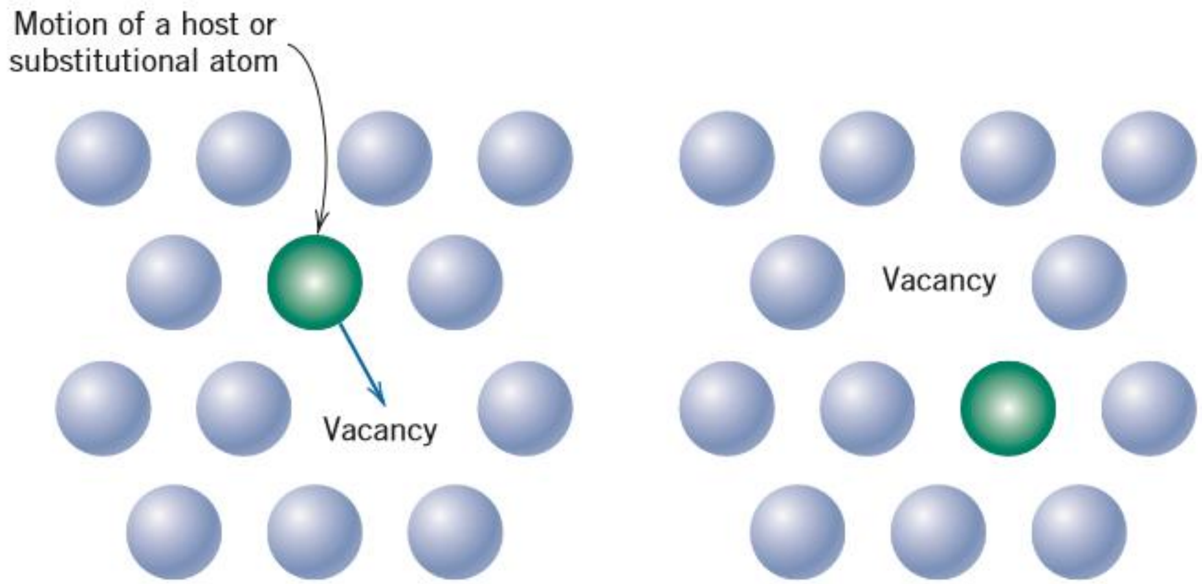
- Diffusion happens by the migration of atoms from lattice site to another lattice site.

- In order for an atom to make such a motion, two conditions are required:
  1) There must be an empty adjacent site.
  2) The atom must have sufficient energy to break bonds with its neighbor atoms and then cause some lattice distortion during the displacement.

- Two diffusion mechanisms:
  1) Vacancy Diffusion
  2) Interstitial Diffusion

Vacancy Diffusion

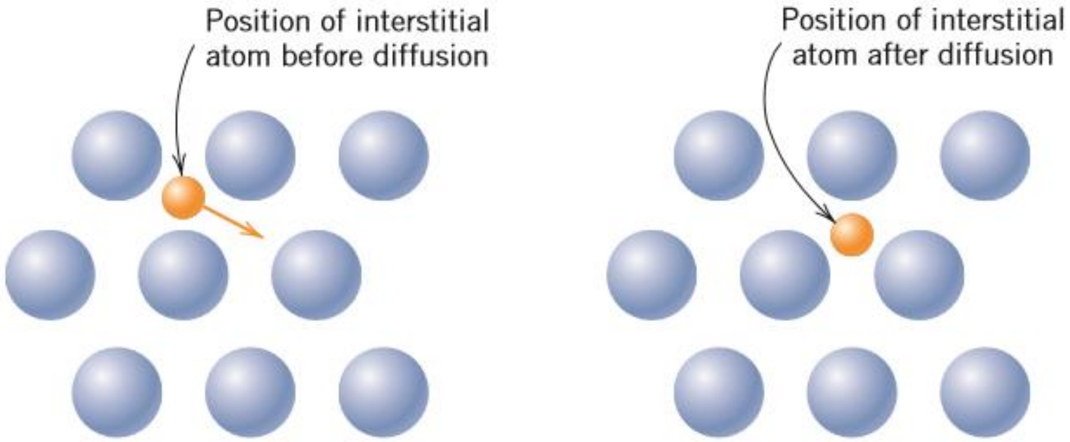
- The **vacancy diffusion mechanism** involves the interchange of an atom from a normal lattice position to an adjacent vacant lattice site. This process necessitates the presence of vacancies.



- Both self-diffusion and interdiffusion occur by this mechanism.

Interstitial Diffusion

- The **interstitial diffusion** involves atoms that migrate from an interstitial position to a neighboring one that is empty.



- This mechanism is found for interdiffusion of impurities such as hydrogen, carbon, nitrogen, and oxygen, which have atoms that are small enough to fit into the interstitial positions.

- In most metal alloys, interstitial diffusion occurs much more rapidly than vacancy diffusion, because the interstitial atoms are smaller and thus more mobile and also there are more empty interstitial positions than vacancies.

Diffusion Flux

- **<u>Diffusion flux </u>**(**J**) is the mass M or the number of atoms diffusing through and perpendicular to a unit cross-sectional area of solid per unit of time.

$$J = \frac{M}{At}$$

A: area across which diffusion is occurring

t: elapsed diffusion time.

- In differential form:

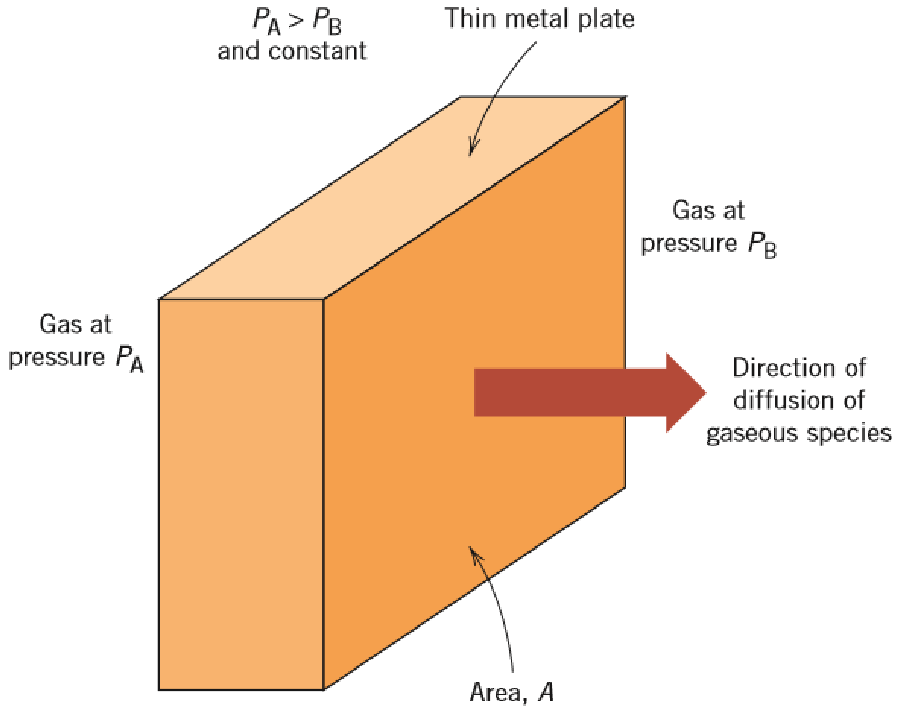
$$J = \frac{1}{A}\frac{dM}{dt}$$

- [J] = $kg/m^2.s$ or $atoms/m^2.s$

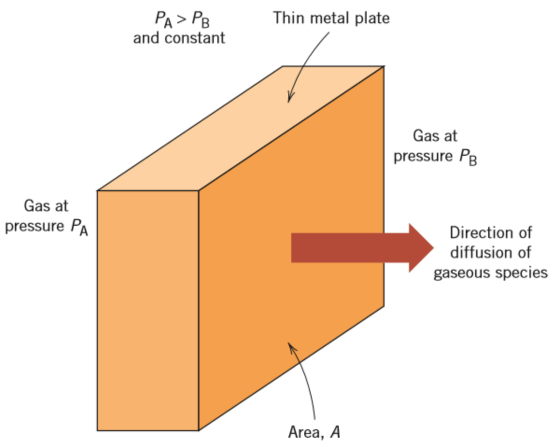
Steady-state Diffusion

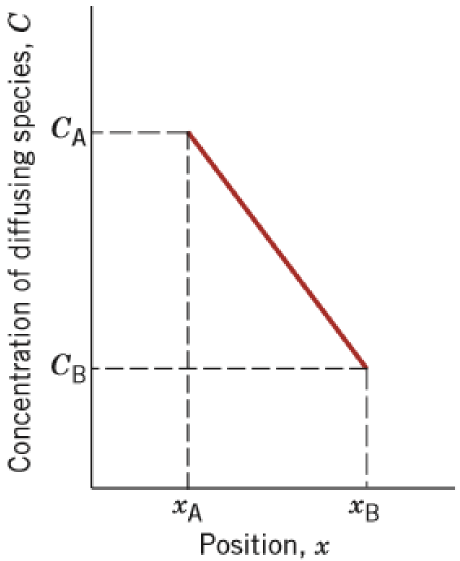
- **Steady-state diffusion** exists when the diffusion flux does not change with time.

- **Example**: diffusion of atoms of a gas through a plate of metal for which the concentrations (or pressures) of the diffusing species on both surfaces of the plate are held constant.



Concentration Profile and Gradient





Concentration Profile

- **<u>Concentration Gradient</u>**: $\frac{dC}{dx}$

- The concentration profile is assumed to be linear.

  ➡ The concentration gradient is: $\frac{\Delta C}{\Delta x} = \frac{C_A - C_B}{x_A - x_B}$

Fick's 1st Law

- **Fick's 1st law**: The diffusion flux is proportional to the concentration gradient for steady state diffusion situations.

$$J = -D\frac{dC}{dx}$$

- C is the concentration and defined as the mass of diffusing species per unit volume of solid (kg/m^3).

- D is called the diffusion coefficient, which is expressed in m^2/s.

- The negative sign indicates that the direction of diffusion is opposite to the concentration gradient.

Example 1

- A plate of iron is exposed to a carburizing (carbon-rich) atmosphere on one side and a decarburizing (carbon-deficient) atmosphere on the other side at $700°C$. If a condition of steady state is achieved, calculate the diffusion flux of carbon through the plate if the concentrations of carbon at positions of 5 and $10\ mm$ beneath the carburizing surface are 1.2 and $0.8\ kg/m^3$, respectively. Assume a diffusion coefficient of $3 \times 10^{-11}\, m^2/s$ at this temperature.

Using Fick's 1st law:

$$J = -D\frac{dC}{dx} = -D\frac{C_A - C_B}{x_A - x_B}$$

$$J = -(3 \times 10^{-11}\, m^2/s)\frac{(1.2 - 0.8)\, kg/m^3}{(5 \times 10^{-3} - 10 \times 10^{-3})m}$$

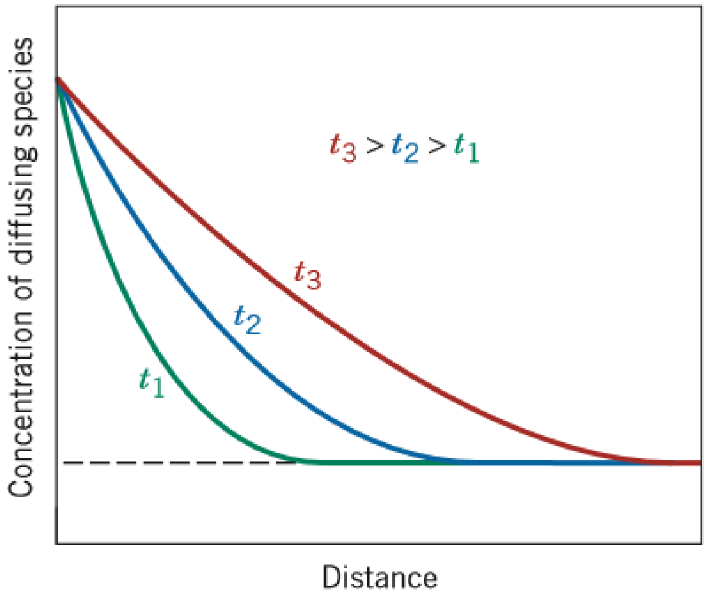
$$J = 2.4 \times 10^{-9}\, kg/m^2.s$$

Fick's 2nd Law

- **<u>Fick's 2nd law</u>**: The time rate of change of concentration is proportional to the second derivative of concentration for non-steady state diffusion situations.

$$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$$

- Concentration profiles for non-steady state diffusion taken at three different times, t_1, t_2, and t_3.



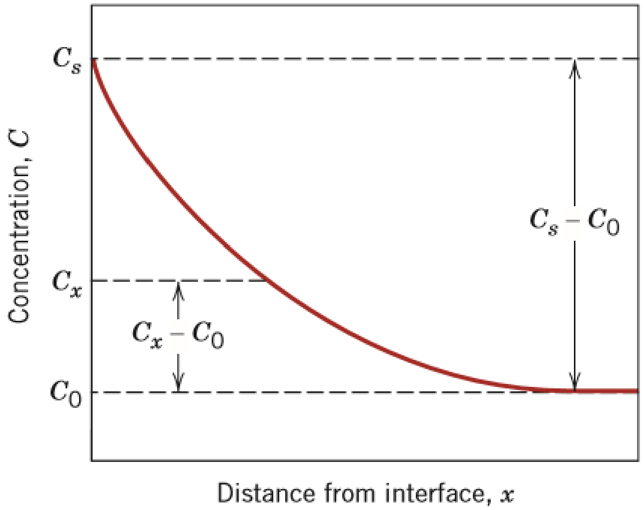
Fick's 2nd Law

- For a constant surface concentration (for a semi-infinite solid), the solution of **Fick's 2nd law**:

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- Where:

C_x: concentration at depth x after time t.
C_s: constant surface concentration at $x = 0$.



- For $t = 0$:

$$C = C_0 \text{ at } 0 \le x \le \infty$$

- For $t > 0$:

$$C = C_s \text{ at } x = 0$$
$$C = C_0 \text{ at } x = \infty$$

- The expression $\operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$ is the Gaussian error function. The values of this function are given for various values of $\frac{x}{2\sqrt{Dt}}$ are given in the next slide.

Fick's 2nd Law

- Values of error function:

z	$erf(z)$	z	$erf(z)$	z	$erf(z)$
0	0	0.55	0.5633	1.3	0.9340
0.025	0.0282	0.60	0.6039	1.4	0.9523
0.05	0.0564	0.65	0.6420	1.5	0.9661
0.10	0.1125	0.70	0.6778	1.6	0.9763
0.15	0.1680	0.75	0.7112	1.7	0.9838
0.20	0.2227	0.80	0.7421	1.8	0.9891
0.25	0.2763	0.85	0.7707	1.9	0.9928
0.30	0.3286	0.90	0.7970	2.0	0.9953
0.35	0.3794	0.95	0.8209	2.2	0.9981
0.40	0.4284	1.0	0.8427	2.4	0.9993
0.45	0.4755	1.1	0.8802	2.6	0.9998
0.50	0.5205	1.2	0.9103	2.8	0.9999

Fick's 2nd Law

- Suppose that it is desired to achieve some specific concentration of solute, C_1, in an alloy:

$$\frac{C_1 - C_0}{C_s - C_0} = cte$$

- Then,

$$\frac{x}{2\sqrt{Dt}} = cte$$

Or

$$\frac{x^2}{DT} = cte$$

Example 2

- Consider an alloy that initially has a uniform carbon concentration of $0.25\ wt\%$ and is to be treated at $950°C$. If the concentration of carbon at the surface is suddenly brought to and maintained at $1.2\ wt\%$,how long will it take to achieve a carbon content of $0.8\ wt\%$ at a position $0.5\ mm$ below the surface? The diffusion coefficient for carbon in iron at this temperature is $1.6 \times 10^{-11}\ m^2/s$; assume that the steel piece is semi-infinite.

$C_0 = 0.25\ wt\%,\ C_s = 1.2\ wt\%,\ C_x = 0.8\ wt\%$ at $x = 0.5mm = 0.5 \times 10^{-3}m$

Using Fick's 2nd law: $\frac{C_x - C_0}{C_s - C_0} = 1 - \mathrm{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$

Let $z = \frac{x}{2\sqrt{Dt}}$: $\frac{(0.8-0.25)wt\%}{(1.2-0.25)wt\%} = 1 - \mathrm{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = 1 - \mathrm{erf}(z)$=> $\mathrm{erf}(z) = 0.421$

z	$\mathrm{erf}(z)$
0.35	0.3794
$z=?$	0.421
0.4	0.4284

By interpolation: $\frac{z-0.35}{0.4-0.35} = \frac{0.421-0.3794}{0.4284-0.3794}$

=> $z = 0.392 = \frac{x}{2\sqrt{Dt}} = \frac{0.5\times10^{-3}m}{2\sqrt{(1.6\times10^{-11}m^2/s)t}}$

=> $t = 25421sec = 7.06\ hrs$

Example 3

- The diffusion coefficients for copper in aluminum at 500 and $600°C$ are 4.8×10^{-14} and $5.3 \times 10^{-13}\, m^2/s$, respectively. Determine the approximate time at $500°C$ that will produce the same diffusion result (in terms of concentration of Cu at some specific point in Al) as a $10\ hrs$ heat treatment at $600°C$.

The composition in both diffusion situations will have the same concentrations and be equal at the same position at both 500 and $600°C$.

=> $C_{500} = C_{600}$ and $x_{500} = x_{600}$

$$\frac{C_{500} - C_0}{C_s - C_0} = \frac{C_{600} - C_0}{C_s - C_0} \qquad \Rightarrow \quad \frac{{x_{500}}^2}{D_{500}t_{500}} = \frac{{x_{600}}^2}{D_{600}t_{600}}$$

=> $D_{500}t_{500} = D_{600}t_{600}$ =>$(4.8 \times 10^{-14}\, m^2/s)t_{500} = (5.3 \times 10^{-13}\, m^2/s)(10\ hrs)$

=>$t_{500} = 110.42\ hrs$

Factors of Diffusion

The factors that influence the diffusion are:

1) **Diffusing Species**

2) **Temperature**

Factors of Diffusion

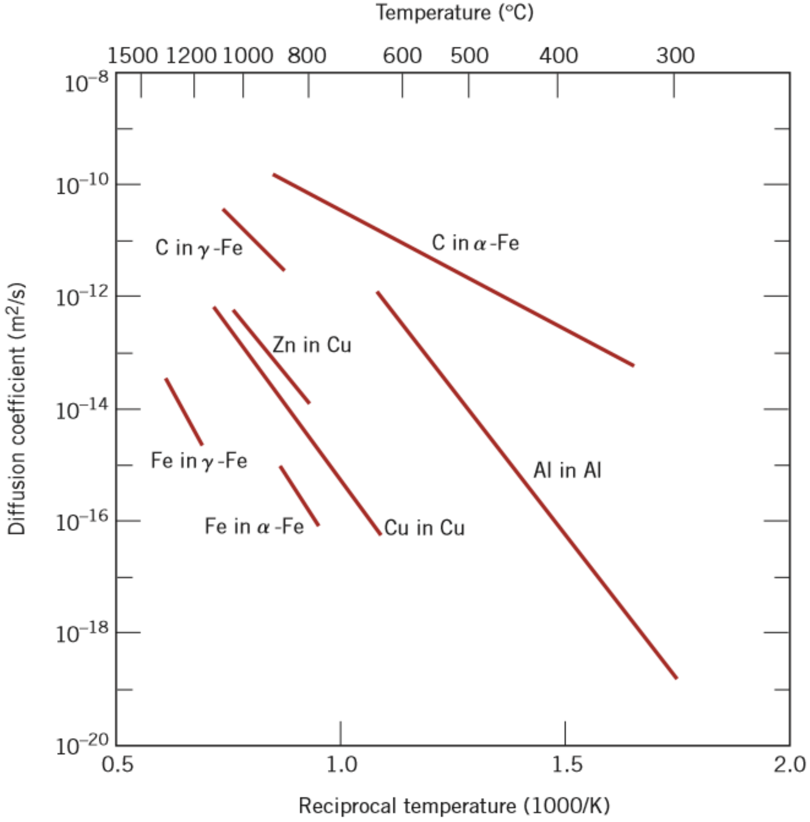
- The magnitude of the diffusion coefficient D indicates the rate at which atoms diffuse and depends on both host and diffusing species as well as on temperature.

Diffusing Species	*Host Metal*	$D_0(m^2/s)$	*Activation Energy Q_d* kJ/mol	*Activation Energy Q_d* eV/atom	*Calculated Value* $T(°C)$	*Calculated Value* $D(m^2/s)$
Fe	α-Fe (BCC)	2.8×10^{-4}	251	2.60	500 900	3.0×10^{-21} 1.8×10^{-15}
Fe	γ-Fe (FCC)	5.0×10^{-5}	284	2.94	900 1100	1.1×10^{-17} 7.8×10^{-16}
C	α-Fe	6.2×10^{-7}	80	0.83	500 900	2.4×10^{-12} 1.7×10^{-10}
C	γ-Fe	2.3×10^{-5}	148	1.53	900 1100	5.9×10^{-12} 5.3×10^{-11}
Cu	Cu	7.8×10^{-5}	211	2.19	500	4.2×10^{-19}
Zn	Cu	2.4×10^{-5}	189	1.96	500	4.0×10^{-18}
Al	Al	2.3×10^{-4}	144	1.49	500	4.2×10^{-14}
Cu	Al	6.5×10^{-5}	136	1.41	500	4.1×10^{-14}
Mg	Al	1.2×10^{-4}	131	1.35	500	1.9×10^{-13}
Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

Factors of Diffusion

- The **diffusion coefficient *D*** is as is a function of temperature.
- The **diffusion coefficient *D***:

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$



D_0 : constant independent of the temperature

Q_d : activation energy for diffusion

R : ideal gas constant = $8.31\, J/mol.k$

T : absolute temperature

Example 4

❖ Compute the diffusion coefficient for magnesium in aluminum at $550°C$.

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

Diffusing Species	*Host Metal*	$D_0(m^2/s)$	*Activation Energy Q_d*		*Calculated Value*	
			kJ/mol	*eV/atom*	$T(°C)$	$D(m^2/s)$
Fe	α-Fe (BCC)	2.8×10^{-4}	251	2.60	500 900	3.0×10^{-21} 1.8×10^{-15}
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Cu	Ni	2.7×10^{-5}	256	2.65	500	1.3×10^{-22}

Example 4

- Compute the diffusion coefficient for magnesium in aluminum at at $550°C$.

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right)$$

Using the table:

$D_0 = 1.2 \times 10^{-4}\, m^2/s$

$Q_d = 131\ KJ/mol\ = 131000\ J/mol$

$T = 550°C + 273 = 823K$

$R = 8.31\, J/mol.k$

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right) = 1.2 \times 10^{-4}\, m^2/s \times \exp\left(-\frac{131000\ J/mol}{8.31\, J/mol.k \times 823K}\right)$$

$D = 5.76 \times 10^{-13}\, m^2/s$

Thank you for your attention!